

MH3100 Real Analysis I

Midterm Practice 3 (Solutions)

AY 2026/2027, Semester 1 – QRS original

Quantitative Research Society @NTU

September 29, 2026

TIME ALLOWED: 60 minutes

Instructions to Students

1. This test contains **FOUR (4)** questions.
2. Answer **all** questions. The marks for each question are indicated below.
3. Write your answers in the space provided. If you need more space, continue on the facing page and clearly indicate that you have done so.
4. This is a **CLOSED BOOK** test.
5. You may use calculators. You may cite any results proved in the lectures and tutorials, **unless you are asked by the question to prove the result.**

Question	1	2	3	4	Total
Total Marks	6	6	6	6	24
Awarded Marks					

Question 1

(6 Marks)

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks) If A and B are nonempty subsets of $\mathbb{R}$ with $A \subseteq B$ and B bounded above, then $\sup A \leq \sup B$.
- (b) (2 marks) The set $A = \{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < 2\}$ has a supremum in $\mathbb{Q}$.
- (c) (2 marks) Every bounded monotone sequence of rational numbers converges to a rational number.

Solution

- (a) True

Proof. Since B is nonempty and bounded above, the Axiom of Completeness asserts that B has a supremum $\sup B$. By the definition of $\sup B$ being an upper bound for B , every element of B is no greater than $\sup B$. Since $A \subseteq B$, every element of A is also an element of B , so every element of A is no greater than $\sup B$; i.e., $\sup B$ is an upper bound for A . Since A is a nonempty subset of $\mathbb{R}$ that is bounded above (by $\sup B$), the Axiom of Completeness also asserts that A has a supremum $\sup A$. The definition of $\sup A$ being the *least* upper bound for A then implies that $\sup A$ is no greater than any upper bound for A , and in particular $\sup A \leq \sup B$. $\square$

- (b) False

Idea. (*Scratch work – not part of the written answer.*) We show that no rational number can be the supremum of $A := \{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < 2\}$ in $\mathbb{Q}$: any rational upper bound s of A must satisfy $s^2 > 2$ (otherwise s is not even an upper bound), yet for such s a strictly smaller rational upper bound $s - h$ can always be produced, so no rational upper bound is least. We also use the classical fact that no rational number squares to 2.

Proof. First, no rational number s satisfies $s^2 = 2$. If $s = p/n$ in lowest terms (so p and n share no common factor greater than 1) satisfied $s^2 = 2$, then $p^2 = 2n^2$, so p^2 is even, hence p is even. Writing $p = 2p'$ then gives $4p'^2 = 2n^2$, so $n^2 = 2p'^2$ is even, hence n is even too, contradicting that p and n share no common factor greater than 1. Therefore $s^2 \neq 2$ for every $s \in \mathbb{Q}$.

Now suppose, towards a contradiction, that $s = \sup_{\mathbb{Q}} A$ exists in $\mathbb{Q}$. Since $1 \in A$ (as $1 > 0$ and $1^2 = 1 < 2$) and s is an upper bound for A , we have $s \geq 1 > 0$.

Suppose $s^2 < 2$. By the Archimedean Property, choose a natural number n satisfying

$$\frac{1}{n} < \min \left\{ 1, \frac{2 - s^2}{2s + 1} \right\},$$

and set $h := 1/n \in \mathbb{Q}$, so $0 < h < 1$ and $h(2s + 1) < 2 - s^2$. Then

$$(s + h)^2 = s^2 + 2sh + h^2 < s^2 + 2sh + h = s^2 + h(2s + 1) < s^2 + (2 - s^2) = 2,$$

using $h^2 < h$ since $0 < h < 1$. Since $s + h \in \mathbb{Q}$, $s + h > 0$, and $(s + h)^2 < 2$, we have $s + h \in A$. But $s + h > s$, contradicting that s is an upper bound for A . Hence $s^2 < 2$ is impossible, and together with $s^2 \neq 2$ we conclude $s^2 > 2$.

Since $s^2 > 2$, by the Archimedean Property choose a natural number n satisfying

$$\frac{1}{n} < \min \left\{ s, \frac{s^2 - 2}{2s} \right\},$$

and set $h := 1/n \in \mathbb{Q}$, so $0 < h < s$ and $2sh < s^2 - 2$. Then $s - h > 0$, and

$$(s - h)^2 = s^2 - 2sh + h^2 > s^2 - 2sh > s^2 - (s^2 - 2) = 2.$$

For every $q \in A$, $q > 0$ and $q^2 < 2 < (s - h)^2$; since q and $s - h$ are both positive, this forces $q < s - h$. Hence $s - h$ is an upper bound for A that is strictly less than s , contradicting that s is the *least* upper bound for A .

Both $s^2 < 2$ and $s^2 > 2$ are impossible, and $s^2 = 2$ is impossible for $s \in \mathbb{Q}$. Therefore no rational number s can be the supremum of A in $\mathbb{Q}$; i.e., A has no supremum in $\mathbb{Q}$. $\square$

(c) False

Idea. (*Scratch work – not part of the written answer.*) Truncating the decimal expansion of $\sqrt{2}$ to n places gives a sequence of rational numbers that is increasing and bounded above by 2, yet converges to the irrational number $\sqrt{2}$.

Proof. For each $n \in \mathbb{N}$, let $S_n := \{k \in \mathbb{N}_0 : k \leq \sqrt{2} \cdot 10^n\}$. Then $0 \in S_n$, so S_n is nonempty, and every $k \in S_n$ satisfies $k < 2 \cdot 10^n$, so S_n is finite. Therefore S_n has a greatest element k_n , and we let $a_n := k_n/10^n \in \mathbb{Q}$ (the truncation of the decimal expansion of $\sqrt{2}$ to n decimal places). Since $k_n \in S_n$ and $k_n + 1 \notin S_n$, we have $k_n \leq \sqrt{2} \cdot 10^n < k_n + 1$, so

$$0 \leq \sqrt{2} - a_n < \frac{1}{10^n} \quad \text{for every } n \in \mathbb{N}. \quad (*)$$

The sequence $\{a_n\}$ is increasing: for each $n \in \mathbb{N}$, we have $10k_n \leq 10 \cdot \sqrt{2} \cdot 10^n = \sqrt{2} \cdot 10^{n+1}$, so $10k_n \in S_{n+1}$; hence $10k_n \leq k_{n+1}$, and dividing by 10^{n+1} gives $a_n \leq a_{n+1}$. It is also bounded above, since $(*)$ gives $a_n \leq \sqrt{2} < 2$ for every $n \in \mathbb{N}$. Thus $\{a_n\}$ is a bounded, increasing (hence monotone) sequence of rational numbers.

We show $\sqrt{2} = \lim a_n$. Let $\varepsilon > 0$ be given. By the Archimedean Property, there is a natural number N satisfying $1/N < \varepsilon$. For every natural number $n \geq N$, $(*)$ and $10^n \geq n$ give

$$\left| a_n - \sqrt{2} \right| = \sqrt{2} - a_n < \frac{1}{10^n} \leq \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

This is exactly the definition of $\sqrt{2} = \lim a_n$. As shown in part (b), no rational number squares to 2, so $\sqrt{2} \notin \mathbb{Q}$. Therefore $\{a_n\}$ is a bounded monotone sequence of rational numbers converging to an irrational number, so the statement is false. $\square$

Marking scheme (6 marks)

QRS suggested marking scheme.

- (2 marks) Part (a): correct answer True.
- (2 marks) Part (b): correct answer False.

- **(2 marks)** Part (c): correct answer False.

No justification is required for full marks on this question; the proofs above are provided for the solutions build only.

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Question 2

(6 Marks)

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks) Every infinite and bounded subset of $\mathbb{R}$ has a limit point.
- (b) (2 marks) The closure of a bounded subset of $\mathbb{R}$ is (sequentially) compact.
- (c) (2 marks) Every set with no limit points is (sequentially) compact.

Solution

- (a) True

Idea. (*Scratch work – not part of the written answer.*) Choose a sequence of distinct points from the infinite bounded set; apply the Bolzano-Weierstrass Theorem to extract a convergent subsequence; since the terms are distinct, all but at most one term of the subsequence lies away from the limit, which forces the limit to be a limit point of the original set.

Proof. Let $S \subseteq \mathbb{R}$ be an infinite and bounded set. Since S is infinite, we may choose a sequence $\{a_n\}$ of pairwise distinct terms in S : having chosen distinct $a_1, \dots, a_{n-1}$, the set $S \setminus \{a_1, \dots, a_{n-1}\}$ is still infinite (a finite set removed from an infinite set leaves an infinite set), hence nonempty, so a further term a_n distinct from all previous ones can always be chosen. Since S is bounded, $\{a_n\}$ is a bounded sequence, and the Bolzano-Weierstrass Theorem asserts that $\{a_n\}$ has a convergent subsequence $\{a_{n_k}\}$; let a denote its limit.

Let $\varepsilon > 0$ be given. Since $a = \lim_{k \rightarrow \infty} a_{n_k}$, there is a natural number K such that every natural number $k \geq K$ satisfies $|a_{n_k} - a| < \varepsilon$. Because the terms a_{n_k} are pairwise distinct, at most one value of k can satisfy $a_{n_k} = a$; hence there is some $k \geq K$ with $a_{n_k} \neq a$ and $|a_{n_k} - a| < \varepsilon$. This term a_{n_k} belongs to $S \setminus \{a\}$ and satisfies $|a_{n_k} - a| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, this is exactly the definition of a being a limit point of S . $\square$

- (b) True

Idea. (*Scratch work – not part of the written answer.*) A closure is always closed; the closure of a bounded set stays inside the same bounding interval, hence stays bounded; a closed and bounded set is compact.

Proof. Let $A \subseteq \mathbb{R}$ be bounded, so there is some $M > 0$ with $|a| \leq M$ for every $a \in A$, i.e., $-M \leq a \leq M$ for every $a \in A$.

We first show $\overline{A}$ is bounded. Let $x \in \overline{A}$ be arbitrary. By the Sequential Criterion for Closures, there is a sequence $\{a_n\}$ in A with $x = \lim a_n$. Since $-M \leq a_n \leq M$ for every $n \in \mathbb{N}$, the Order Limit Theorem (applied to $\{a_n\}$ against the constant sequences $-M$ and M) gives $-M \leq x \leq M$, i.e., $|x| \leq M$. Since $x \in \overline{A}$ was arbitrary, every element of $\overline{A}$ satisfies $|x| \leq M$, so $\overline{A}$ is bounded.

Next, $\overline{A}$ is closed. By the theorem on Limit Points of Closures, a number is a limit point of $\overline{A}$ if and only if it is a limit point of A ; every limit point of A lies in $\overline{A}$ by the definition of closure. Therefore every limit point of $\overline{A}$ lies in $\overline{A}$, so $\overline{A}$ contains all its own limit points, i.e., $\overline{A}$ is closed.

Since $\overline{A}$ is closed and bounded, the Sequential Compactness of Closed and Bounded Sets asserts that $\overline{A}$ is (sequentially) compact. $\square$

(c) False

Idea. (*Scratch work – not part of the written answer.*) Take $A = \mathbb{N}$. Consecutive natural numbers are at least 1 apart, so $\mathbb{N}$ has no limit points, yet $\mathbb{N}$ is unbounded and so cannot be compact.

Proof. Consider $A = \mathbb{N} = \{1, 2, 3, \dots\}$. We first show A has no limit points. Suppose, towards a contradiction, that some $a \in \mathbb{R}$ is a limit point of A . Applying the definition of limit point with $\varepsilon = 1/2$ gives some $x_1 \in A \setminus \{a\}$ with $|x_1 - a| < 1/2$; in particular $|x_1 - a| > 0$, since $x_1 \neq a$. Applying the definition again, this time with $\varepsilon = |x_1 - a| > 0$, gives some $x_2 \in A \setminus \{a\}$ with $|x_2 - a| < |x_1 - a|$. Since $|x_2 - a| < |x_1 - a|$, we have $x_2 \neq x_1$. Now x_1 and x_2 are distinct elements of $A = \mathbb{N}$, so $|x_1 - x_2| \geq 1$ (distinct natural numbers differ by at least 1). But the triangle inequality gives

$$|x_1 - x_2| \leq |x_1 - a| + |a - x_2| < |x_1 - a| + |x_1 - a| = 2|x_1 - a| < 2 \cdot \frac{1}{2} = 1,$$

contradicting $|x_1 - x_2| \geq 1$. Therefore no real number a is a limit point of A ; i.e., A has no limit points.

Next, A is unbounded: by the Archimedean Property, for every real number M there is a natural number n with $n > M$, so no real number is an upper bound for A . Consequently A is not (sequentially) compact: the sequence $a_n := n$ is contained in A , and every subsequence $\{a_{n_k}\}$ satisfies $a_{n_k} = n_k \geq k$ for every $k \in \mathbb{N}$ (since the indices of a subsequence are strictly increasing), so $\{a_{n_k}\}$ is itself unbounded. By the contrapositive of the Boundedness of Convergent Sequences, an unbounded sequence cannot converge, so $\{a_{n_k}\}$ does not converge. Since every subsequence of $\{a_n\}$ diverges, A is not (sequentially) compact.

Thus $A = \mathbb{N}$ has no limit points yet is not compact, so the statement is false. $\square$

Marking scheme (6 marks)

QRS suggested marking scheme.

- (2 marks) Part (a): correct answer True.
- (2 marks) Part (b): correct answer True.
- (2 marks) Part (c): correct answer False.

No justification is required for full marks on this question; the proofs above are provided for the solutions build only.

Question 3

(6 Marks)

In this question, A and B are sets of real numbers. **Write down the definition** of a limit point of a set, and **use the definition to show** that if x is a limit point of $A \cup B$, then x is a limit point of A or x is a limit point of B . Do not use the Sequential Criterion for Limit Points.

Solution

Definition (Limit Point). A number $a \in \mathbb{R}$ is a limit point of a set $S \subseteq \mathbb{R}$ if for every $\varepsilon > 0$, there exists $x \in S \setminus \{a\}$ such that $|x - a| < \varepsilon$.

Method 1: Contrapositive

Idea. (*Scratch work – not part of the written answer.*) Suppose x is a limit point of neither A nor B . Negating the definition twice gives two positive radii $\varepsilon_1, \varepsilon_2$ that separately keep A and B away from x . Taking $\varepsilon = \min\{\varepsilon_1, \varepsilon_2\}$ keeps $A \cup B$ away from x too, so x is not a limit point of $A \cup B$.

Proof. We prove the contrapositive: if x is a limit point of neither A nor B , then x is not a limit point of $A \cup B$.

Suppose x is not a limit point of A . Negating the definition of a limit point, there exists a number $\varepsilon_1 > 0$ such that no $y \in A \setminus \{x\}$ satisfies $|y - x| < \varepsilon_1$; equivalently, every $y \in A$ with $|y - x| < \varepsilon_1$ is equal to x .

Suppose likewise that x is not a limit point of B . There exists a number $\varepsilon_2 > 0$ such that every $z \in B$ with $|z - x| < \varepsilon_2$ is equal to x .

Let $\varepsilon := \min\{\varepsilon_1, \varepsilon_2\}$, which is positive since both ε_1 and ε_2 are positive. Let $w \in A \cup B$ satisfy $|w - x| < \varepsilon$. Then $w \in A$ or $w \in B$. If $w \in A$, then since $|w - x| < \varepsilon \leq \varepsilon_1$, the choice of ε_1 forces $w = x$. If instead $w \in B$, then since $|w - x| < \varepsilon \leq \varepsilon_2$, the choice of ε_2 forces $w = x$. In either case, $w = x$. Therefore every $w \in A \cup B$ with $|w - x| < \varepsilon$ is equal to x ; equivalently, no $w \in (A \cup B) \setminus \{x\}$ satisfies $|w - x| < \varepsilon$. This is exactly the negation of the definition of x being a limit point of $A \cup B$, so x is not a limit point of $A \cup B$.

We have shown that if x is a limit point of neither A nor B , then x is not a limit point of $A \cup B$. Taking the contrapositive, if x is a limit point of $A \cup B$, then x is a limit point of A or x is a limit point of B . $\square$

Common slip. A common error is taking $\varepsilon := \varepsilon_1 + \varepsilon_2$ or $\varepsilon := \max\{\varepsilon_1, \varepsilon_2\}$ instead of $\varepsilon := \min\{\varepsilon_1, \varepsilon_2\}$; the larger radius may exceed the radius that was shown to work for one of the two sets, so the key steps “ $\varepsilon \leq \varepsilon_1$ ” and “ $\varepsilon \leq \varepsilon_2$ ” fail, and the argument breaks down. Another common error is proving only that x being a limit point of $A \cup B$ implies x is a limit point of A (forgetting the “or” and the symmetric role of B) by carelessly working with a fixed one of the two sets throughout the proof.

Marking scheme (6 marks)

QRS suggested marking scheme.

- **(2 marks)** Correct definition of a limit point, with the quantifiers in the correct order (“for every $\varepsilon > 0$ ” before “there exists $x \in S \setminus \{a\}$ ”) and the deleted-point set $S \setminus \{a\}$ present. 1 mark if the quantifier order is reversed or $S \setminus \{a\}$ is written as S .

- **(3 marks)** Key steps of the proof: 1 mark for correctly negating the definition of limit point for A (obtaining ε_1); 1 mark for the same for B (obtaining ε_2); 1 mark for taking $\varepsilon = \min\{\varepsilon_1, \varepsilon_2\}$ and correctly concluding that no point of $(A \cup B) \setminus \{x\}$ lies within ε of x .
- **(1 mark)** The three writing expectations: complete sentences, statements joined by connectives, and inference shown with “therefore”/“hence”/“thus”/“so” or $\therefore$. Deduct this mark for a proof written as a bare symbolic chain.

Typical deductions: using the Sequential Criterion for Limit Points despite the prohibition (0 marks for the proof); forgetting to exclude x itself when forming $A \setminus \{x\}$ or $B \setminus \{x\}$; taking $\varepsilon = \varepsilon_1 + \varepsilon_2$ or $\max\{\varepsilon_1, \varepsilon_2\}$ instead of the minimum; asserting the contrapositive conclusion without proving it.

Question 4

(6 Marks)

Write down the definition of a (sequentially) compact set, and **use the definition to show** that if K_1 and K_2 are compact sets, then $K_1 \cup K_2$ is compact. *Do not use the Sequential Compactness of Closed and Bounded Sets or the Bolzano-Weierstrass Theorem.*

Solution

Definition ((Sequentially) Compact Set). A set $K \subseteq \mathbb{R}$ is said to be (sequentially) compact if every sequence contained in K has a convergent subsequence whose limit lies in K .

Method 1: Direct construction of a convergent subsequence in $K_1 \cup K_2$

Idea. (*Scratch work – not part of the written answer.*) Given a sequence in $K_1 \cup K_2$, infinitely many of its indices must have terms lying in K_1 , or infinitely many must have terms lying in K_2 (otherwise both index sets would be finite, and their union could not be all of $\mathbb{N}$). Whichever set catches infinitely many terms provides a subsequence to which that set's own compactness can be applied.

Proof. Let $\{a_n\}$ be an arbitrary sequence contained in $K_1 \cup K_2$. For each $n \in \mathbb{N}$, assign a_n to K_1 if $a_n \in K_1$, and otherwise assign a_n to K_2 (so $a_n \in K_2$ in this case); let $S_1 := \{n \in \mathbb{N} : a_n \text{ was assigned to } K_1\}$ and $S_2 := \{n \in \mathbb{N} : a_n \text{ was assigned to } K_2\}$. Then S_1 and S_2 partition $\mathbb{N}$, so $S_1 \cup S_2 = \mathbb{N}$. Since the union of two finite sets is finite while $\mathbb{N}$ is infinite, at least one of S_1, S_2 must be infinite; without loss of generality, say S_1 is infinite.

List the elements of the infinite set $S_1 \subseteq \mathbb{N}$ in strictly increasing order as $n_1 < n_2 < n_3 < \dots$; this listing can be continued indefinitely, since an infinite subset of $\mathbb{N}$ has no greatest element. Then $\{a_{n_k}\}_{k=1}^\infty$ is a subsequence of $\{a_n\}$, and by the choice of S_1 , every term a_{n_k} lies in K_1 .

Since K_1 is compact and $\{a_{n_k}\}$ is a sequence contained in K_1 , the definition of compactness applied to K_1 asserts that $\{a_{n_k}\}$ has a convergent subsequence $\{a_{n_{k_j}}\}_{j=1}^\infty$ whose limit a lies in K_1 . The index maps $k \mapsto n_k$ and $j \mapsto k_j$ are both strictly increasing (being index maps of subsequences), so their composition $j \mapsto n_{k_j}$ is also strictly increasing; hence $\{a_{n_{k_j}}\}$ is a subsequence of the original sequence $\{a_n\}$. This subsequence converges to $a \in K_1 \subseteq K_1 \cup K_2$.

Since $\{a_n\}$ was an arbitrary sequence contained in $K_1 \cup K_2$, and it has been shown to have a convergent subsequence with limit in $K_1 \cup K_2$, the definition of compactness asserts that $K_1 \cup K_2$ is (sequentially) compact. $\square$

Common slip. Students often assume without proof that a sequence in $K_1 \cup K_2$ must lie entirely in K_1 or entirely in K_2 ; this is false in general, since the sequence may alternate between the two sets. The correct argument instead shows only that infinitely many indices land in one of the two sets, which suffices to build a subsequence lying entirely in that set. Another common error is extracting a convergent subsequence of $\{a_{n_k}\}$ but forgetting to verify that the resulting index sequence n_{k_j} is still strictly increasing in j , which is exactly what makes it a genuine subsequence of the original sequence $\{a_n\}$.

Marking scheme (6 marks)

QRS suggested marking scheme.

- **(2 marks)** Correct definition of a (sequentially) compact set: every sequence in K has a convergent subsequence with limit in K . 1 mark if “in K ” is dropped from either the subsequence's convergence or its limit.

- **(3 marks)** Key steps: 1 mark for correctly splitting the index set $\mathbb{N}$ into two pieces and arguing at least one is infinite; 1 mark for constructing the subsequence $\{a_{n_k}\}$ lying entirely in one K_i with strictly increasing indices; 1 mark for correctly applying the compactness of that K_i to extract the final convergent subsequence and noting it is still a subsequence of $\{a_n\}$.
- **(1 mark)** The three writing expectations: complete sentences, connectives, and explicit inference (“therefore”/“hence”/“thus”/“so” or $\therefore$).

Typical deductions: citing the Sequential Compactness of Closed and Bounded Sets or the Bolzano-Weierstrass Theorem (0 marks for the proof, since both are prohibited); asserting without justification that “infinitely many terms lie in K_1 or K_2 ”; forgetting to check that the extracted sub-subsequence is a genuine subsequence of the original $\{a_n\}$; omitting “without loss of generality” when only one case is argued.